

Unit-2

Series Solutions of Ordinary Differential Equations

Let $y = \sum_{n=0}^{\infty} C_n x^n$

$$= C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + \dots$$

$$y' = \sum_{n=1}^{\infty} n C_n x^{n-1}$$

$$= C_1 + 2C_2 x + 3C_3 x^2 + 4C_4 x^3 + \dots$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2}$$

$$= 2C_2 + 6C_3 x + 12C_4 x^2 + \dots$$

Power series expressions of elementary functions;

$$(1) e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$(2) \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \\ = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

$$(3) \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \\ = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$(4) \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

$$(5) \sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$(6) \ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

$$(7) \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$= 1 + x + x^2 + x^3 + \dots$$

$$(8) (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3 + \dots$$

* Linear Differential Eqn.

$$A(x)y'' + B(x)y' + C(x)y = 0$$

$$\Rightarrow y'' + P(x)y' + Q(x)y = 0$$

$$\Rightarrow P(x) = B(x)/A(x), \quad Q(x) = C(x)/A(x)$$

(Q) Find a particular solution of the given differential equation $y'' + 3y' + 4y = 4e^x$ by using the method of variation of parameters.

Solution
 $y'' + 3y' + 4y = 4e^x$

The homo - eqn, is

$$y'' + 3y' + 4y = 0$$

The C.E is

$$r^2 + 3r + 4 = 0$$

$$(r+1)(r+4) = 0$$

$$r_1 = -1, r_2 = -4$$

$\therefore$ The fundamental solutions are

$$y_1 = e^{-x}, y_2 = e^{-4x}$$

$\therefore$ The homo - solution is

$$y_h = c_1 e^{-x} + c_2 e^{-4x}$$

The Wronskian of y_1 and y_2 is

$$W(y_1, y_2) = \begin{vmatrix} e^{-x} & e^{-4x} \\ -e^{-x} & -4e^{-4x} \end{vmatrix}$$

$$= (e^{-x})(-4e^{-4x}) - (-e^{-5x})(e^{-4x})$$

$$= -4e^{-5x} + e^{-5x}$$

$$= -3e^{-5x}$$

The P - s is

$$y_p = u_1 y_1 + u_2 y_2$$

Let $f(x) = 4e^x$

$$u_1 = - \int \frac{y_2 f(x)}{W} dx$$

$$= - \int \frac{e^{-4x} \cdot 4e^x}{-e^{-5x}} dx$$

$$= \int \frac{4e^{-3x}}{e^{-5x}} dx = \int 4e^{2x} dx = 2e^{2x}$$

$$u_2 = \int \frac{y_1 f(x)}{W} dx$$

$$= \int \frac{e^{-x} \cdot 4e^x}{-e^{-5x}} dx$$

$$= \int \frac{4}{-e^{-3x}} dx = \int -4e^{3x} dx = -\frac{4}{3} e^{3x}$$

$$\therefore y_p = (2e^{2x})(e^{-x}) + (-\frac{4}{3}e^{3x})(e^{-4x})$$

$$= 2e^x - \frac{4}{3}e^{-x}$$

$$= \frac{2}{3}e^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$r = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$$

$$= \frac{-(-2) \pm \sqrt{(-2)^2 + 4(1)(1)}}{2(1)}$$

$$= \frac{2 \pm \sqrt{-4}}{2}$$

$$= \frac{2 \pm 2i}{2} = 1 \pm i$$

$\therefore$ The C-S is

$$y_c(x) = e^x (c_1 \cos x + c_2 \sin x)$$

$$\text{Let } g(x) = x + 1$$

$$y_p = Ax + B$$

$$y_p' = A$$

$$y_p'' = 0$$

$$y_p'' - 2y_p' + 2y_p = g(x)$$

$$0 - 2(A) + 2(Ax + B) = x + 1$$

$$(2A)x + (2B - 2A) = x + 1$$

$$2A = 1 \Rightarrow A = \frac{1}{2}$$

$$2B - 2A = 1 \Rightarrow B = 1$$

$$\therefore y_p(x) = \frac{1}{2}x + 1$$

$\therefore$ The g-s is

$$y(x) = y_c(x) + y_p(x)$$

$$= e^x (c_1 \cos x + c_2 \sin x) + \frac{1}{2}x + 1$$

$$y'(x) = e^x (c_1 \cos x + c_2 \sin x) + e^x (-c_1 \sin x + c_2 \cos x) + \frac{1}{2}$$

$$y(0) = 3$$

$$e^0 (c_1 \cos 0 + c_2 \sin 0) + \frac{1}{2}(0) + 1 = 3$$

$$c_1 + 1 = 3$$

$$c_1 = 2$$

$$y'(0) = 0$$

$$e^0 (c_1 \cos 0 + c_2 \sin 0) + e^0 (-c_1 \sin 0 + c_2 \cos 0) + \frac{1}{2} = 0$$

$$c_1 + c_2 + \frac{1}{2} = 0$$

$$2 + c_2 + \frac{1}{2} = 0$$

$$c_2 = -\frac{5}{2}$$

$\therefore$ The required solution is

$$y(x) = e^x (2 \cos x - \frac{5}{2} \sin x) + \frac{1}{2}x + 1$$

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∴ The C-S is

$$y_c(x) = e^{-3x} (C_1 \cos 2x + C_2 \sin 2x).$$

Let $g(x) = e^{-3x} \cdot \cos 2x$.

∴ $y_p(x) = e^{-3x} (Ax \cos 2x + Bx \sin 2x)$

(7) Find the general form of a particular solution of $y^{(3)} + ay' = x \sin x + x^2 e^{2x}$.

solution

$$y^{(3)} + ay' = x \sin x + x^2 e^{2x}$$

The homo-egen, is

$$y^{(3)} + ay' = 0$$

The C-E is

$$r^3 + ar = 0$$

$$r(r^2 + a) = 0$$

$$r = 0 \text{ (or) } r^2 + a = 0$$

$$r = \pm 3i$$

∴ The C-S is

$$y_c(x) = C_1 + C_2 \cos 3x + C_3 \sin 3x.$$

Let $g(x) = x \sin x + x^2 e^{2x}$

$$g_1(x) = x \sin x$$

$$y_{p1}(x) = (Ax + B) \cos x + (Cx + D) \sin x$$

$$g_2(x) = x^2 e^{2x}$$

$$y_{p2}(x) = e^{2x} (Ex^2 + Fx + G)$$

∴ The P-S is

$$y_p = y_{p1} + y_{p2}$$

$$= (Ax + B) \cos x + (Cx + D) \sin x + e^{2x} (Ex^2 + Fx + G).$$

(8) Solve the initial value problem

$$y'' - ay' + ay = x + 1; y(0) = 3, y'(0) = 0$$

solution

$$y'' - ay' + ay = x + 1; y(0) = 3, y'(0) = 0$$

The homo-egen, is

$$y'' - ay' + ay = 0$$

The C-E is

$$r^2 - ar + a = 0$$

$$a=1, b=-2, c=2$$

5) Find a particular solution of $y'' + 4y = 3x^3$.

Solution

$$y'' + 4y = 3x^3$$

The homogeneous equation is

$$y'' + 4y = 0$$

The C.E is

$$r^2 + 4 = 0$$

$$r^2 = -4$$

$$r = \pm 2i$$

$$r = \alpha \pm \beta i$$

$$r = 0 \pm 2i$$

$$\alpha = 0, \beta = 2$$

$\therefore$ The complementary solution is

$$y_c(x) = C_1 \cos 2x + C_2 \sin 2x.$$

$$\text{Let } g(x) = 3x^3$$

$$y_p = Ax^3 + Bx^2 + Cx + D$$

$$y_p' = 3Ax^2 + 2Bx + C$$

$$y_p'' = 6Ax + 2B$$

$$y_p'' + 4y_p = g(x)$$

$$(6Ax + 2B) + 4(Ax^3 + Bx^2 + Cx + D) = 3x^3$$

$$6Ax + 2B + 4Ax^3 + 4Bx^2 + 4Cx + 4D = 3x^3$$

$$(4A)x^3 + (4B)x^2 + (6A + 4C)x + (2B + 4D) = 3x^3$$

$$4A = 3 \Rightarrow A = 3/4$$

$$4B = 0 \Rightarrow B = 0$$

$$4A + 4C = 0 \Rightarrow C = -9/8$$

$$2B + 4D = 0 \Rightarrow D = 0.$$

$\therefore$ The required particular solution is

$$y_p = 3/4 x^3 - 9/8 x.$$

6) Determine the appropriate form for a particular solution is $y'' + 6y' + 13y = e^{-3x} \cos 2x$.

Solution:

$$y'' + 6y' + 13y = e^{-3x} \cos 2x$$

The homo - eqn is

$$y'' + 6y' + 13y = 0$$

The C.E is

$$r^2 + 6r + 13 = 0$$

$$a = 1, b = 6, c = 13$$

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-6 \pm \sqrt{6^2 - 4(1)(13)}}{2(1)}$$

$$= \frac{-6 \pm \sqrt{-16}}{2}$$

$$= \frac{-6 \pm 4i}{2} = -3 \pm 2i$$

$$(2) \quad y(3) - y'' - 5y' - 2y = 0$$

The C.E. is characteristic equation

$$2x^3 - x^2 - 5x - 2 = 0$$

$$(x-2)(2x^2 + 3x + 1) = 0$$

$$(x-2)(2x+1)(x+1) = 0$$

$$\therefore x_1 = 2, x_2 = -1/2, x_3 = -1$$

$\therefore$ The g.s. is

$$y(x) = c_1 e^{2x} + c_2 e^{-1/2 x} + c_3 e^{-x}$$

$$(3) \quad y(4) + 18y'' + 81y = 0$$

The C.E. is

$$x^4 + 18x^2 + 81 = 0$$

$$(x^2 + 9)^2 = 0$$

$$x^2 = -9 \quad x = \pm 3i$$

$$x = \pm 3i \quad x = \pm 3i$$

$$x_1 = 3i, x_2 = -3i, x_3 = 3i, x_4 = -3i$$

The g.s. is

$$y(x) = c_1 \cos 3x + c_2 x \cos 3x$$

$$+ c_3 \sin 3x + c_4 x \sin 3x$$

$$= c_1 + c_2 x \cos 3x + (c_3 + c_4 x) \sin 3x$$

(4) The given $y_1 = e^x, y_2 = e^{2x}$ and $y_3 = e^{3x}$ are linearly independent solutions of the third order homogeneous equation $y^3 - 6y'' + 11y' - 6y = 0$. Find a particular solution of the given equation that satisfies the initial conditions $y(0) \neq 0, y'(0) = 0, y''(0) = 3$

Solution

$$y^3 - 6y'' + 11y' - 6y = 0$$

$y_1 = e^x, y_2 = e^{2x}$ and $y_3 = e^{3x}$ are the solutions of the given differential equation.

$$y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$$

$$y' = c_1 e^x + 2c_2 e^{2x} + 3c_3 e^{3x}$$

$$y'' = c_1 e^x + 4c_2 e^{2x} + 9c_3 e^{3x}$$

when

$$y(0) = 0 \Rightarrow c_1 + c_2 + c_3 = 0$$

$$y'(0) = 0 \Rightarrow c_1 + 2c_2 + 3c_3 = 0$$

$$y''(0) = 3 \Rightarrow c_1 + 4c_2 + 9c_3 = 3$$

By solving

$$c_1 = 3/2, c_2 = -3, c_3 = 3/2$$

$\therefore$ The particular solution is

$$y(x) = 3/2 e^x - 3e^{2x} + 3/2 e^{3x}$$

$$g(x) = \sin ax \text{ (or) } \cos ax$$

$$\Rightarrow y_p = A \cos(ax) + B \sin(ax)$$

Combination Form

function of trigonometric (if you can) or combination of both forms if you are allowed.

eg: $x^2 e^{2x}$ (or) $x \sin x$

Unit.1Linear Equations of Higher OrderDistinct root

$$r_1 \neq r_2 \Rightarrow y_h = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

Repeated root

$$r_1 = r_2 = r \Rightarrow y_h = c_1 e^{rx} + c_2 x e^{rx} \\ = (c_1 + c_2 x) e^{rx}$$

Complex root

$$r = \alpha \pm \beta i \Rightarrow y_h = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

Polynomial Form

$$g(x) = x \text{ or } \cos x \text{ or } \sin x$$

$$\Rightarrow y_p = x \text{ or polynomial if}$$

*eg: $g(x) = x^2$

$$y_p = Ax^2 + Bx + C$$

Exponential form

$$g(x) = e^{ax}$$

$$\Rightarrow y_p = A e^{ax}$$

(1) Find the general solution of the differential equation. $9y^{(3)} + 12y'' + 4y' = 0$.

Solution.

$$9y^{(3)} + 12y'' + 4y' = 0$$

The characteristic equation is

$$9r^3 + 12r^2 + 4r = 0$$

$$r^3 + \frac{12}{9}r^2 + \frac{4}{9}r = 0$$

$$r(r^2 + \frac{12}{9}r + \frac{4}{9}) = 0$$

$$r(r + \frac{2}{3})(r + \frac{2}{3}) = 0$$

$$\therefore r_1 = 0, r_2 = -\frac{2}{3}, r_3 = -\frac{2}{3}$$

$\therefore$ The general solution is

$$y(x) = c_1 + c_2 e^{-\frac{2}{3}x} + c_3 x e^{-\frac{2}{3}x}$$

Series Solutions of Ordinary Differential Equations

$$\text{Let } y = \sum_{n=0}^{\infty} c_n x^n$$

$$= c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$$

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$$= c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

$$= 2c_2 + 6c_3 x + 12c_4 x^2 + \dots$$

Power series expressions of elementary functions;

$$(1) e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$(2) \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$(3) \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

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$$(4) \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

$$(5) \sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$(6) \ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$(7) \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$= 1 + x + x^2 + x^3 + \dots$$

$$(8) (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3 + \dots$$

* Linear Differential Eqn/

$$A(x)y'' + B(x)y' + C(x)y = 0$$

$$\Rightarrow y'' + p(x)y' + q(x)y = 0$$

$$\Rightarrow p(x) = B(x)/A(x) \quad q(x) = C(x)/A(x)$$

log 4 $\Rightarrow$ 8
log 3 $\Rightarrow$ 8
log 4 $\Rightarrow$ 8

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(1) solve the equation $y' + xy = 0$ by using power series method.

Solution

$$y' + xy = 0$$

$$\text{let } y = \sum_{n=0}^{\infty} c_n x^n = \dots$$

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1} = \dots$$

substituting in the given equation,

$$\sum_{n=1}^{\infty} n c_n x^{n-1} + 2 \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+1) c_{n+1} x^n + 2 \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+1) c_{n+1} + 2 c_n] x^n = 0$$

The identity principle,

$$(n+1) c_{n+1} + 2 c_n = 0$$

$$c_{n+1} = -\frac{2 c_n}{(n+1)}, \quad n \geq 0$$

$$\text{for } n=0, c_1 = -\frac{2 c_0}{1} = -2 c_0$$

$$n=1, c_2 = -\frac{2 c_1}{2} = -\frac{2(-2 c_0)}{2} = \frac{2^2 c_0}{2!}$$

$$n=2, c_3 = -\frac{2 c_2}{3} = -\frac{2(2^2 c_0)}{3!} = -\frac{2^3 c_0}{3!}$$

$$n=3, c_4 = -\frac{2 c_3}{4} = -\frac{2(-2^3 c_0)}{4 \times 3!} = \frac{2^4 c_0}{4!}$$

⋮

$$c_n = \frac{(-2)^n c_0}{n!}$$

∴ The required power series solution is

$$\begin{aligned} y(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n 2^n c_0 x^n}{n!} \\ &= c_0 \sum_{n=0}^{\infty} \frac{(-2x)^n}{n!} \\ &= c_0 e^{-2x} \end{aligned}$$

(2) solve the equation $y'' + y = 0$ by using power series method.

Solution

$$y'' + y = 0$$

$$\text{let } y = \sum_{n=0}^{\infty} c_n x^n = \dots$$

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1} = \dots$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} = \dots$$

Substitute in the given equation,

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2} x^n + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)c_{n+2} + c_n] x^n = 0$$

The identity principle,

$$(n+2)(n+1)c_{n+2} + c_n = 0$$

$$c_{n+2} = -\frac{c_n}{(n+2)(n+1)}, \quad n \geq 0$$

$$\text{For } n=0, \quad c_2 = -\frac{c_0}{(2)(1)} = -\frac{c_0}{2!}$$

$$n=1, \quad c_3 = -\frac{c_1}{(3)(2)} = -\frac{c_1}{3!}$$

$$n=2, \quad c_4 = -\frac{c_2}{(4)(3)} = \frac{-(-c_0)}{4 \times 3 \times 2!} = \frac{c_0}{4!}$$

$$n=3, \quad c_5 = -\frac{c_3}{(5)(4)} = -\frac{(-c_1)}{5 \times 4 \times 3!} = \frac{c_1}{5!}$$

$$n=4, \quad c_6 = -\frac{c_4}{(6)(5)} = -\frac{c_0}{6 \times 5 \times 4!} = -\frac{c_0}{6!}$$

$$n=5, \quad c_7 = -\frac{c_5}{(7)(6)} = -\frac{(-c_1)}{7 \times 6 \times 5!} = \frac{c_1}{7!}, \dots$$

$\therefore$ The required power series solution is

$$y(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + c_7 x^7 + \dots$$

$$= c_0 + c_1 x + \left(-\frac{c_0}{2!}\right) x^2 + \left(-\frac{c_1}{3!}\right) x^3 + \left(\frac{c_0}{4!}\right) x^4$$

$$+ \left(\frac{c_1}{5!}\right) x^5 + \left(-\frac{c_0}{6!}\right) x^6 + \left(-\frac{c_1}{7!}\right) x^7 + \dots$$

$$= c_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right)$$

$$+ c_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots\right)$$

$$= c_0 \cos x + c_1 \sin x.$$

Self-study

$$(3) \quad y' = y \Rightarrow y = c_0 e^x.$$

$$(4) \quad y' = 4y \Rightarrow y = c_0 e^{4x}.$$

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(5) Solve the equation $(x-3)y' + 2y = 0$ by using power series method. Determine the radius of convergence of the resulting series.

Solution:

$$(x-3)y' + 2y = 0$$

$$\text{Let } y = \sum_{n=0}^{\infty} c_n x^n = \dots$$

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1} = \dots$$

Substitute into the given equation,

$$(x-3) \sum_{n=1}^{\infty} n c_n x^{n-1} + 2 \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=1}^{\infty} n c_n x^n - 3 \sum_{n=1}^{\infty} n c_n x^{n-1} + 2 \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=0}^{\infty} n c_n x^n - 3 \sum_{n=0}^{\infty} (n+1) c_{n+1} x^n + 2 \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=0}^{\infty} [n c_n - 3(n+1) c_{n+1} + 2 c_n] x^n = 0$$

The identity principle,

$$n c_n - 3(n+1) c_{n+1} + 2 c_n = 0$$

$$c_{n+1} = \frac{n+2}{3(n+1)} c_n, \quad n \geq 0.$$

$$\text{For } n=0, \quad c_1 = \frac{2}{3} c_0$$

$$n=1, \quad c_2 = \frac{3}{3(2)} c_1 = \frac{3}{3^2} c_0$$

$$n=2, \quad c_3 = \frac{4}{3(3)} c_2 = \frac{4}{3^3} c_0$$

$\vdots$

$$c_n = \frac{n+1}{3^n} c_0, \quad \text{if } n \geq 1.$$

$\therefore$ The power series solution is

$$y(x) = c_0 \sum_{n=0}^{\infty} \frac{n+1}{3^n} x^n.$$

Its radius of convergence is

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \frac{3n+3}{n+2}$$

$$= \lim_{n \rightarrow \infty} \frac{3n/n + 3/n}{n/n + 2/n} = \lim_{n \rightarrow \infty} \frac{3 + 3/n}{1 + 2/n} = \frac{3+0}{1+0}$$

$$= 3.$$

(6) $(x-1)y' + 2y = 0$

* $\sum_{n=0}^{\infty} [n c_n - (n+1) c_{n+1} + 2 c_n] x^n = 0$

* $c_{n+1} = \frac{n+2}{n+1} c_n, \quad n \geq 0$

* $y(x) = \sum_{n=0}^{\infty} (n+1) x^n$

* $\rho = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{n+2} = 1$

(7) Solve the initial value problem

$(x^2+1)y'' + 2xy' - 2y = 0; \quad y(0) = 0, y'(0) = 1$

by using power series method.

Solution:

$(x^2+1)y'' + 2xy' - 2y = 0; \quad y(0) = 0, y'(0) = 1$

Let $y = \sum_{n=0}^{\infty} c_n x^n = \dots$

$y' = \sum_{n=1}^{\infty} n c_n x^{n-1} = \dots$

$y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} = \dots$

Substituting in the given equation,

$(x^2+1) \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + 2x \sum_{n=1}^{\infty} n c_n x^{n-1} - 2 \sum_{n=0}^{\infty} c_n x^n = 0$

$-2 \sum_{n=0}^{\infty} c_n x^n = 0$

$\sum_{n=2}^{\infty} n(n-1) c_n x^n + \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$

$+ 2 \sum_{n=1}^{\infty} n c_n x^n - 2 \sum_{n=0}^{\infty} c_n x^n = 0$

$\sum_{n=0}^{\infty} n(n-1) c_n x^n + \sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n$

$+ 2 \sum_{n=0}^{\infty} n c_n x^n - 2 \sum_{n=0}^{\infty} c_n x^n = 0$

$\sum_{n=0}^{\infty} [(n+2)(n+1) c_{n+2} + n(n-1) c_n + 2n c_n - 2 c_n] x^n = 0$

$\sum_{n=0}^{\infty} [(n+2)(n+1) c_{n+2} + (n+2)(n-1) c_n] x^n = 0$

The identity principle,

$(n+2)(n+1) c_{n+2} + (n+2)(n-1) c_n = 0$

$c_{n+2} = \frac{(1-n)}{(1+n)} c_n, \quad n \geq 0$

for $n=0, c_2 = c_0$

$n=1, c_3 = 0$

$n=2, c_4 = -\frac{1}{3}c_2 = -\frac{1}{3}c_0$

$n=3, c_5 = -\frac{1}{2}c_3 = 0$

$n=4, c_6 = -\frac{3}{5}c_4 = \frac{1}{5}c_0$

$n=5, c_7 = -\frac{2}{3}c_5 = 0, \dots$

$\therefore$ The general solution is

$$\begin{aligned} y &= c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + \dots \\ &= c_0 + c_1x + c_0x^2 + (-\frac{1}{3}c_0)x^4 + (\frac{1}{5}c_0)x^6 + \dots \\ &= c_1x + c_0[1 + x^2 - \frac{x^4}{3} + \frac{x^6}{5} + \dots] \\ y' &= c_1 + c_0[2x - 4\frac{x^3}{3} + 6\frac{x^5}{5} + \dots] \end{aligned}$$

when

$$y(0) = 0 \Rightarrow c_0 = 0$$

$$y'(0) = 1 \Rightarrow c_1 = 0$$

$\therefore$ The required solution is

$$y(x) = x$$

$$(8) y'' - 4y = 0; y(0) = 2, y'(0) = 0$$

$$(9) y'' + 4y = 0; y(0) = 0, y'(0) = 3$$

*

$$(8) c_0 = 2, c_1 = 0$$

$$c_{n+2} = \frac{4}{(n+2)(n+1)} c_n, n \geq 0$$

$$y = 2 \left[1 + \frac{(x)^2}{2!} + \frac{(x)^4}{4!} + \frac{(x)^6}{6!} + \dots \right]$$

$$\sum_{n=0}^{\infty} \frac{(x)^{2n}}{(2n)!} = \cosh(x)$$

$$= 2 \cosh x$$

*

$$(9) c_0 = 0, c_1 = 3$$

$$c_{n+2} = \frac{-4}{(n+2)(n+1)} c_n, n \geq 0$$

$$y = \frac{3}{2} \left[(x) - \frac{(x)^3}{3!} + \frac{(x)^5}{5!} - \frac{(x)^7}{7!} + \dots \right]$$

$$= \frac{3}{2} \sin(x)$$

(1) Find the general solution in power of x of $(x^2-4)y'' + 3xy' + y = 0$. Find the particular solution with $y(0) = 4, y'(0) = 1$.

$$(x^2-4)y'' + 3xy' + y = 0$$

$$y'' + \frac{3x}{x^2-4}y' + \frac{1}{x^2-4}y = 0$$

Compare with $y'' + P(x)y' + Q(x)y = 0$

$$P(x) = \frac{3x}{x^2-4}, Q(x) = \frac{1}{x^2-4}$$

$$x^2-4 = 0 \Rightarrow x = \pm 2$$

$\therefore$ The singular points $x = \pm 2$.

$$\text{Let } y(x) = \sum_{n=0}^{\infty} c_n x^n$$

$$= \dots$$

$$y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$$= \dots$$

$$y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

$$= \dots$$

Substituting in the given equation,

$$(x^2-4) \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

$$+ 3x \sum_{n=1}^{\infty} n c_n x^{n-1} + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - 4 \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + 3x \sum_{n=1}^{\infty} n c_n x^{n-1} + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) c_n x^n - 4 \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + 3 \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) c_n x^n - 4 \sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n + 3 \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=0}^{\infty} [n(n-1) c_n - 4(n+2)(n+1) c_{n+2} + 3n c_n + c_n] x^n = 0$$

The identity principle,

$$n(n-1) c_n - 4(n+2)(n+1) c_{n+2} + 3n c_n + c_n = 0$$

$$(n(n-1) + 3n + 1) c_n - (4(n+2)(n+1)) c_{n+2} = 0$$

$$(n^2 + 2n + 1) c_n - (4(n+2)(n+1)) c_{n+2} = 0$$

$$(n+1)^2 c_n - (4(n+2)(n+1)) c_{n+2} = 0$$

$$c_{n+2} = \frac{(n+1)^2}{4(n+2)(n+1)} c_n$$

$$= \frac{n+1}{4(n+2)} c_n, \text{ for } n \geq 0$$

$$\text{For } n=0 \Rightarrow c_2 = \frac{1}{4(n)} c_0 = \frac{1}{8} c_0$$

$$n=1 \Rightarrow c_3 = \frac{2}{4(3)} c_1 = \frac{1}{6} c_1$$

$$n=2 \Rightarrow c_4 = \frac{3}{4(4)} c_2 = \frac{3}{16} \left(\frac{1}{8} c_0 \right) = \frac{3}{128} c_0$$

$$n=3 \Rightarrow c_5 = \frac{4}{4(5)} c_3 = \frac{1}{5} \left(\frac{1}{6} c_1 \right) = \frac{1}{30} c_1, \dots$$

The general solution is

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + \dots$$

$$= c_0 + c_1 x + \left(\frac{1}{8} c_0 \right) x^2 + \left(\frac{1}{6} c_1 \right) x^3 + \left(\frac{3}{128} c_0 \right) x^4$$

$$+ \left(\frac{1}{30} c_1 \right) x^5 + \dots$$

$$= c_0 \left(1 + \frac{1}{8} x^2 + \frac{3}{128} x^4 + \dots \right)$$

$$+ c_1 \left(x + \frac{1}{6} x^3 + \frac{1}{30} x^5 + \dots \right)$$

$$y' = c_0 \left(\frac{1}{4} x + \frac{3}{32} x^3 + \dots \right)$$

$$+ c_1 \left(1 + \frac{1}{2} x^2 + \frac{1}{6} x^4 + \dots \right)$$

$$\text{when } y(0) = 4 \Rightarrow c_0 = 4$$

$$y'(0) = 1 \Rightarrow c_1 = 1$$

$$\begin{aligned} y(x) &= 4 \left(1 + \frac{1}{8} x^2 + \frac{3}{128} x^4 + \dots \right) \\ &+ 1 \left(x + \frac{1}{6} x^3 + \frac{1}{30} x^5 + \dots \right) \\ &= \left(4 + \frac{1}{2} x^2 + \frac{3}{32} x^4 + \dots \right) \\ &+ \left(x + \frac{1}{6} x^3 + \frac{1}{30} x^5 + \dots \right) \\ &= 4 + x + \frac{1}{2} x^2 + \frac{1}{6} x^3 + \frac{3}{32} x^4 + \frac{1}{30} x^5 + \dots \end{aligned}$$

H.W

(c) Solve the initial value problem

$$y'' + xy' - xy = 0; \quad y(0) = 1, \quad y'(0) = 0$$

- The identity principle

$$c_{n+2} c_{n+1} = c_{n+2} + c_{n-2} c_n = 0$$

$$c_{n+2} = \frac{-c_{n-2}}{c_{n+2} c_{n+1}} c_n, \quad \text{for } n \geq 0$$

$$- y(0) = 1 \Rightarrow c_0 = 1$$

$$y'(0) = 0 \Rightarrow c_1 = 0$$

$$- y(x) = 1 + x^2$$

(3) Find the general solution in power of x of the differential equation

$$(x^2+2)y'' + 4xy' + 2y = 0.$$

Solution

$$(x^2+2)y'' + 4xy' + 2y = 0$$

$$y'' + \frac{4x}{x^2+2} y' + \frac{2}{x^2+2} y = 0$$

compare with $y'' + P(x)y' + Q(x)y = 0$

$$P(x) = \frac{4x}{x^2+2}, \quad Q(x) = \frac{2}{x^2+2}$$

$$x^2+2=0 \Rightarrow x = \pm \sqrt{2}i$$

$\therefore$ The singular points $x = \pm \sqrt{2}i$

The identity principle,

$$2(n+2)(n+1)(n+2) + (n+1)(n+2) c_n = 0$$

$$c_{n+2} = -\frac{(n+1)(n+2)}{2(n+2)(n+1)} c_n = -\frac{1}{2} c_n, \text{ for } n \geq 0$$

$$\begin{aligned} * y(x) &= c_0 \left(1 - \frac{1}{2} x^2 + \frac{1}{4} x^4 - \frac{1}{8} x^6 + \dots \right) \\ &\quad + c_1 \left(x - \frac{1}{2} x^3 + \frac{1}{4} x^5 - \frac{1}{8} x^7 + \dots \right) \end{aligned}$$

(4) Find the general solution in power of x of the differential equation

$$(x^2-1)y'' - 6xy' + 2y = 0$$

Solution

$$(x^2-1)y'' - 6xy' + 2y = 0$$

$$y'' - \frac{6x}{x^2-1} y' + \frac{2}{x^2-1} y = 0$$

compare with $y'' + P(x)y' + Q(x)y = 0$

$$P(x) = -\frac{6x}{x^2-1}, \quad Q(x) = \frac{2}{x^2-1}$$

$$x^2-1=0 \Rightarrow x = \pm 1$$

$\therefore$ The singular points $x = \pm 1$

The identity principle,

$$-(n+2)(n+1)(n+2) + (n^2-7n+2) c_n = 0$$

$$c_{n+2} = \frac{n^2-7n+2}{(n+2)(n+1)} c_n, \text{ for } n \geq 0$$

$$\begin{aligned} * y(x) &= c_0 \left(1 + x^2 - \frac{7}{3} x^4 + \frac{2}{9} x^6 + \dots \right) \\ &\quad + c_1 \left(x - \frac{7}{3} x^3 + \frac{1}{3} x^5 - \frac{4}{63} x^7 + \dots \right) \end{aligned}$$

15) Solve the initial value problem

$$(t^2 - 2t - 3) \frac{d^2 y}{dt^2} + 3(t-1) \frac{dy}{dt} + y = 0;$$

$y(1) = 4, y'(1) = -1$, by using power series method.

Solution.

$$(t^2 - 2t - 3) \frac{d^2 y}{dt^2} + 3(t-1) \frac{dy}{dt} + y = 0$$

$$\text{let } x = t-1 \Rightarrow t = x+1$$

$$\begin{aligned} t^2 - 2t - 3 &= (x+1)^2 - 2(x+1) - 3 \\ &= x^2 + 2x + 1 - 2x - 2 - 3 \\ &= x^2 - 4 \end{aligned}$$

$$3(t-1) = 3x$$

substituting in the given equation,

$$(x^2 - 4) y'' + 3xy' + y = 0$$

$$* c_{n+2} = \frac{n+1}{4(n+2)} c_n, \text{ for } n \geq 0$$

$$\begin{aligned} * y(x) &= 4 - x + \frac{1}{2} x^2 - \frac{1}{6} x^3 + \frac{3}{32} x^4 - \frac{1}{30} x^5 + \dots \\ \therefore y(t) &= 4 - (t-1) + \frac{1}{2} (t-1)^2 - \frac{1}{6} (t-1)^3 \\ &\quad + \frac{3}{32} (t-1)^4 - \frac{1}{30} (t-1)^5 + \dots \end{aligned}$$

16) Find the regular singular point of the differential equation

$$x^2(1+x)y'' + x(4-x^2)y' + (2+3x)y = 0$$

solution:

$$x^2(1+x)y'' + x(4-x^2)y' + (2+3x)y = 0$$

$$y'' + \frac{x(4-x^2)}{x^2(1+x)} y' + \frac{(2+3x)}{x^2(1+x)} y = 0$$

compare with $y'' + P(x)y' + Q(x)y = 0$

$$P(x) = \frac{x(4-x^2)}{x^2(1+x)} = \frac{(2-x)(2+x)}{x(1+x)}$$

$$Q(x) = \frac{2+3x}{x^2(1+x)}$$

$$x^2(1+x) = 0 \Rightarrow x = 0 \text{ (or)} x = -1$$

$\therefore$ The singular points are $x = 0$ (or) $x = -1$.

If $x = 0$,

$$P(x) = \frac{(2-x)(2+x)}{x(1+x)}$$

$$= \frac{(2-x)(2+x)}{(1+x)}$$

$$\lim_{x \rightarrow 0} p(x) = \lim_{x \rightarrow 0} \frac{(2-x)(2+x)}{1+x}$$

$$= (2-0)(2+0) = 4 \text{ (finite)}$$

$$Q(x) = \frac{(x-0)^2 Q(x)}{1+x}$$

$$q(x) = (x-0)^2 Q(x) = x^2 \frac{(2+3x)}{1+x} = \frac{2+3x}{1+x}$$

$$\lim_{x \rightarrow 0} q(x) = \lim_{x \rightarrow 0} \frac{2+3x}{1+x} = \frac{2+0}{1+0} = 2 \text{ (finite)}$$

$\therefore x=0$ is regular solution point.

If $x=-1$,

$$p(x) = (x-(-1)) p(x)$$

$$= (x+1) \frac{(2-x)(2+x)}{x(1+x)} = \frac{(2-x)(2+x)}{x}$$

$$\lim_{x \rightarrow -1} p(x) = \lim_{x \rightarrow -1} \frac{(2-x)(2+x)}{x}$$

$$= \frac{(2-(-1))(2+(-1))}{(-1)}$$

$$= \frac{(3)(1)}{(-1)} = -3 \text{ (finite)}$$

$$q(x) = (x-(-1))^2 Q(x)$$

$$= (x+1)^2 \frac{2+3x}{x^2(1+x)} = \frac{(x+1)(2+3x)}{x^2}$$

$$\lim_{x \rightarrow -1} q(x) = \lim_{x \rightarrow -1} \frac{(x+1)(2+3x)}{x^2}$$

$$= \frac{(-1+1)(2+3(-1))}{(-1)^2}$$

$$= \frac{(0)(-1)}{1} = 0 \text{ (finite)}$$

$\therefore x=-1$ is regular singular point.

$\therefore$ The regular singular points of the given equation are $x=0$ and $x=-1$.

Prob (7) Find the regular singular point of the differential equation

$$3x^3 y'' + 2x^2 y' + (1-x^2)y = 0$$

solution:

$$3x^3 y'' + 2x^2 y' + (1-x^2)y = 0$$

$$y'' + \frac{2x^2}{3x^3} y' + \frac{(1-x^2)}{3x^3} y = 0$$

compare with $y'' + p(x)y' + q(x)y = 0$

$$p(x) = \frac{2x^2}{3x^3} = \frac{2}{3x}$$

$$q(x) = \frac{1-x^2}{3x^3}$$

$$3x^3 = 0 \Rightarrow x = 0$$

$\therefore$ The singular point $x = 0$.

$$p(x) = (x-0)p(x) = x \cdot \frac{2}{3x} = \frac{2}{3}$$

$$\lim_{x \rightarrow 0} p(x) = \lim_{x \rightarrow 0} \frac{2}{3} = \frac{2}{3} \text{ (finite)}$$

$$q(x) = (x-0)^2 q(x) = x^2 \cdot \frac{(1-x^2)}{3x^3} = \frac{1-x^2}{3x}$$

$$\lim_{x \rightarrow 0} q(x) = \lim_{x \rightarrow 0} \frac{1-x^2}{3x} = \frac{1-0}{3(0)} = \frac{1}{0} = \infty \text{ (not finite)}$$

$\therefore x=0$ is not regular singular point of the given equation.

Ex 1

(8) Find the regular singular point of the differential equation

$$(1-x^2)y'' + xy' + x^2y = 0.$$

Solution.

$$(1-x)y'' + xy' + x^2y = 0$$

$$y'' + \frac{x}{1-x} y' + \frac{x^2}{1-x} y = 0$$

compare with $y'' + p(x)y' + q(x)y = 0$

$$p(x) = \frac{x}{1-x}, \quad q(x) = \frac{x^2}{1-x}$$

$$1-x=0 \Rightarrow x=1.$$

$\therefore$ The singular point $x=1$.

$$p(x) = (x-1)p(x) = (x-1) \frac{x}{1-x} = -x$$

$$\lim_{x \rightarrow 1} p(x) = \lim_{x \rightarrow 1} -x = -1 \text{ (finite)}$$

$$q(x) = (x-1)^2 q(x) = (x-1)^2 \frac{x^2}{1-x} = -(x-1)x^2$$

$$\lim_{x \rightarrow 1} q(x) = \lim_{x \rightarrow 1} -(x-1)x^2 = -(1-1)(1)^2 = 0 \text{ (finite)}$$

$\therefore x=1$ is the regular singular point of the given equation.